TUTORIAL 10

Null Spaces and Linear Maps

NOTATION AND CONVENTIONS

For $A \in M_{m\times n}(F)$, the homogeneous solution set is denoted by

$$\ker A = \{x \in F^n : Ax = 0\}.$$

Here F^n is the set of columns of length n over the field F, and 0 on the right is the zero column of length m. The rank $\operatorname{rank} A$ is the number of leading entries in its reduced row echelon form. A variable whose column has no leading entry is a free variable.

NOTATION AND CONVENTIONS

If $A \in M_{m\times n}(F)$, then $Ax = b$ is a system of m equations in n variables, with $x \in F^n$ and $b \in F^m$. The notation $[A \mid b]$ denotes its augmented matrix, and $A \sim B$ denotes row equivalence. The rank $\operatorname{rank} A$ is the number of pivots. Define $\ker A = \{x \in F^n : Ax = 0\}$. The expression $\operatorname{span}\{v_1, \ldots, v_k\}$ denotes all combinations $\alpha_1 v_1 + \cdots + \alpha_k v_k$ with $\alpha_j \in F$. The notation $x_0 + U = \{x_0 + u : u \in U\}$ denotes a translate of a subspace U. Distinguish the empty set $\varnothing$, with no elements, from $\{0\}$, which has one element.

MOTIVATION AND GUIDING IDEA

Motivation. In $Ax = b$, the column space determines attainable measurements, while the kernel determines undetectable changes in the state. Distinguishing these roles allows existence and uniqueness to be tested before solving explicitly.

NOTATION AND CONVENTIONS

- $\mathbb{F}^{m\times n}$ is the space of matrices with m rows and n columns over the field $\mathbb{F}$; A^T denotes the transpose.
- $\mathcal{R}(A) = \operatorname{Row}(A) \subseteq \mathbb{F}^n$ is the row space; $\mathcal{C}(A) = \operatorname{Col}(A) \subseteq \mathbb{F}^m$ is the column space. Identify a row with its transposed column when matrix multiplication requires it.
- $N(A) = \ker A = \{x \in \mathbb{F}^n : Ax = 0\}$ is the kernel; $\operatorname{nullity}(A) = \dim \ker A$ is the nullity; $\operatorname{rank}(A) = \dim \operatorname{Col}(A) = \dim \operatorname{Row}(A)$ is the rank.
- $\operatorname{span} S$ consists of all finite linear combinations of vectors in S; e_j is the standard basis vector with entry 1 at index j and 0 elsewhere.
- $x_0 + \ker A = \{x_0 + v : v \in \ker A\}$ is an affine space, a translate of the kernel. Its affine dimension is $\dim \ker A$; it need not be a vector subspace.

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DEFINITION 10.1

Let $A \in \mathbb{F}^{m\times n}$, and define $T_A : \mathbb{F}^n \to \mathbb{F}^m$ by $T_A(x) = Ax$. Then $\operatorname{Im} T_A = \operatorname{Col}(A)$ and $\ker T_A = N(A)$, so

$$\operatorname{rank}(A) + \dim \ker A = n.$$

The right-hand side is the dimension of the **domain**. The system $Ax = b$ is solvable if and only if $b \in \operatorname{Col}(A)$. If x_0 is one solution, **all** solutions form $x_0 + \ker A$. An existing solution is therefore unique if and only if $\ker A = \{0\}$. The nonzero echelon rows form a basis of the row space, but a column-space basis must use the **original** columns at the pivot indices.

MOTIVATION AND GUIDING IDEA

Tutorial objectives. Distinguish the row space, column space and null space, and use rank and nullity to quantify information and freedom in a system. The measurement examples apply to engineering systems in many disciplines.

10.1 The null space and nullity of a matrix

DEFINITION 10.2

Let $\mathbb{F}$ be a field and $A \in \mathbb{F}^{m\times n}$.
The **null space of** A, also called its *kernel*, is the set of column vectors

$$N(A) := \{x \in \mathbb{F}^n \mid Ax = 0\},$$

that solve the homogeneous system $Ax = 0$.

- $N(A)$ is a vector subspace of $\mathbb{F}^n$.
- Its dimension is the **nullity** of A, denoted

$$\dim N(A) = \operatorname{null}(A).$$

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PROPOSITION 10.1

For $A \in \mathbb{F}^{m\times n}$,

$$\operatorname{rank}(A) + \operatorname{null}(A) = n.$$

- A has n columns, each a vector in $\mathbb{F}^m$.
- Reduce A to reduced row echelon form.
- The number of pivot columns equals $\operatorname{rank}(A)$. The columns of the *original matrix* at these pivot indices form a basis of $\operatorname{Col}(A)$; row reduction itself need not preserve the column space.
- There are $n - \operatorname{rank}(A)$ nonpivot columns. These correspond to free variables in $Ax = 0$, each contributing one independent parameter to its solution space. Consequently,

$$\dim N(A) = \operatorname{null}(A) = n - \operatorname{rank}(A).$$

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- Therefore,

$$\operatorname{rank}(A) + \operatorname{null}(A) = \operatorname{rank}(A) + \big(n - \operatorname{rank}(A)\big) = n.$$

EXERCISE 10.1

$$A = \begin{pmatrix} 1 & -3 & 0 \\ 0 & 1 & 2 \end{pmatrix}.$$

Please find a basis and dimension for the kernel, please give every solution of $Ax = 0$, and please check rank-nullity.

WORKED SOLUTION

The first two columns are pivot columns. Every solution arises from $t \in \mathbb{R}$; the nonzero spanning vector is a kernel basis.

$$x_1 - 3x_2 = 0, \quad x_2 + 2x_3 = 0, \quad x_3 = t \implies x_2 = -2t,\ x_1 = -6t.$$

$$\ker A = \operatorname{span}\{(-6, -2, 1)^T\}, \qquad \operatorname{rank} A + \dim \ker A = 2 + 1 = 3.$$

MOTIVATION AND GUIDING IDEA

Different questions require different spaces. The column space describes attainable right-hand sides. The null space describes changes in the input that leave the output unchanged. Rank and nullity measure these two kinds of freedom.

10.2 Interpreting matrix-vector multiplication

EXERCISE 10.2

Let $\mathbb{F}$ be a field and $B \in \mathbb{F}^{n \times m}$. Define

$$J := \{Bv \mid v \in \mathbb{F}^m\} \subseteq \mathbb{F}^n.$$

Please show that

$$J = \operatorname{Col}(B).$$

WORKED SOLUTION

Denote the columns of B by

$$B = \begin{pmatrix} | & & | \\ b_1 & \dots & b_m \\ | & & | \end{pmatrix},$$

so that $b_1, \dots, b_m \in \mathbb{F}^n$ are its column vectors.
Now let

$$v = \begin{pmatrix} \lambda_1 \\ \vdots \\ \lambda_m \end{pmatrix} \in \mathbb{F}^m.$$

By the definition of matrix-vector multiplication,

$$Bv = \lambda_1 b_1 + \cdots + \lambda_m b_m.$$

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EXERCISE 10.2 CONTINUED

We prove both inclusions.

1. First, $J \subseteq \mathrm{Col}(B)$.

   Let $y \in J$. Then $y = Bv$ for some $v \in \mathbb{F}^m$.

   Every such vector is a linear combination of $b_1, \ldots, b_m$:

   $$y = Bv = \lambda_1 b_1 + \cdots + \lambda_m b_m$$

   for suitable scalars $\lambda_1, \ldots, \lambda_m \in \mathbb{F}$.

   Thus y belongs to their span, namely

   $$y \in \mathrm{span}\{b_1, \ldots, b_m\} = \mathrm{Col}(B).$$

   This proves $J \subseteq \mathrm{Col}(B)$.

2. Conversely, $\mathrm{Col}(B) \subseteq J$.

   Let $w \in \mathrm{Col}(B)$. There are scalars $\eta_1, \ldots, \eta_m \in \mathbb{F}$ such that

   $$w = \eta_1 b_1 + \cdots + \eta_m b_m.$$

   Define

   $$v := \begin{pmatrix} \eta_1 \\ \vdots \\ \eta_m \end{pmatrix} \in \mathbb{F}^m.$$

   The preceding calculation gives

   $$Bv = \eta_1 b_1 + \cdots + \eta_m b_m = w.$$

   Hence $w = Bv$ for some $v \in \mathbb{F}^m$, so

   $$w \in J.$$

   This proves $\mathrm{Col}(B) \subseteq J$.

The two inclusions yield

$$J = \mathrm{Col}(B),$$

as required.

EXERCISE 10.3

Let $\mathbb{F}$ be a field and let $A, B \in \mathbb{F}^{n \times n}$ satisfy

$$AB = 0.$$

Please show that

$$\mathrm{Col}(B) \subseteq N(A),$$

Every column of B, and every linear combination of its columns, is therefore annihilated by A.

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EXERCISE 10.3 CONTINUED

WORKED SOLUTION

Use the definitions of column space and null space.
It suffices to prove that $Av = 0$ for every $v \in \mathrm{Col}(B)$.
By the definition of column space, $v \in \mathrm{Col}(B)$ implies that some $w \in \mathbb{F}^n$ satisfies

$$v = Bw.$$

Compute

$$Av = A(Bw) = (AB)w.$$

Since $AB = 0$,

$$(AB)w = 0w = 0,$$

and therefore

$$Av = 0.$$

We have proved that $Av = 0$ for every $v \in \mathrm{Col}(B)$, or equivalently,

$$v \in N(A).$$

Consequently,

$$\mathrm{Col}(B) \subseteq N(A),$$

as required.

EXERCISE 10.4 FOUNDATION

$$A = \begin{pmatrix} 1 & -1 & 0 & 2 \\ 0 & 0 & 1 & 3 \\ 1 & -1 & 1 & 5 \end{pmatrix}, \qquad b = (2, 4, 6)^T.$$

Please find row-space, column-space and kernel bases and please check rank-nullity. Please give a solvability condition for a general right-hand side and please find all solutions for the given b.

WORKED SOLUTION

The first two rows are independent and pivot columns are $1, 3$. The kernel vectors have independent free entries in positions $2, 4$. The row relation gives the solvability condition, and the first two equations prove sufficiency. The displayed solutions are a translate of the kernel.

$$R_3 = R_1 + R_2, \quad B_{\text{row}} = \{(1, -1, 0, 2), (0, 0, 1, 3)\}, \quad \text{rank } A = 2,$$

$$B_{\text{col}} = \{(1, 0, 1)^T, (0, 1, 1)^T\},$$

$$x_2 = s, \quad x_4 = t \quad \Longrightarrow \quad x_1 = s - 2t, \quad x_3 = -3t,$$

$$B_{\text{ker}} = \{(1, 1, 0, 0)^T, (-2, 0, -3, 1)^T\}, \qquad 2 + 2 = 4.$$

$$\exists x : \; Ax = b \iff b_3 = b_1 + b_2,$$

$$x = (2, 0, 4, 0)^T + s(1, 1, 0, 0)^T + t(-2, 0, -3, 1)^T, \qquad s, t \in \mathbb{R}.$$

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MOTIVATION AND GUIDING IDEA

Motivation. The same information may appear as a polynomial, a matrix or a coordinate vector. A linear map compares such descriptions while preserving linear combinations. The kernel asks what is lost; the image asks what can be attained.

NOTATION AND CONVENTIONS

- $T : V \to W$ denotes a map with domain V and codomain W; $T(v) \in W$ is the image of v. The vectors $0_V, 0_W$ are the zero vectors in the corresponding spaces.
- $\ker T = \{v \in V : T(v) = 0_W\}$ is the kernel; $\operatorname{Im} T = T(V) = \{T(v) : v \in V\}$ is the image. Do not automatically identify the image with the codomain.
- $\operatorname{rank} T = \dim \operatorname{Im} T$ is the rank; $\dim \ker T$ is the nullity. The notation $V \cong W$ means that a linear isomorphism between them exists.
- $B = (v_1, \ldots, v_n)$ is an **ordered** basis. If $v = \sum_{j=1}^{n} a_j v_j$, then $[v]_B = (a_1, \ldots, a_n)^T$. The order of basis vectors determines the order of entries.
- $\mathbb{F}_{\leq n}[X]$ is the space of polynomials of degree at most n, including the zero polynomial. For $T : V \to V$, $T^2 = T \circ T$ denotes composition, not entrywise squaring.

DEFINITION **10.3**

Let V, W be vector spaces over the same field $\mathbb{F}$. A map $T : V \to W$ is linear if it is well defined and satisfies

$$T(av + bw) = aT(v) + bT(w) \qquad (v, w \in V,\ a, b \in \mathbb{F}).$$

It is injective if $T(v) = T(w)$ implies $v = w$, and surjective if every $y \in W$ equals $T(v)$ for some $v \in V$. For a linear map, injectivity is equivalent to $\ker T = \{0_V\}$, and surjectivity to $\operatorname{Im} T = W$. An isomorphism is a bijective linear map; its inverse is linear. If $\dim V < \infty$, then

$$\dim V = \dim \ker T + \dim \operatorname{Im} T.$$

The coordinate map $v \mapsto [v]_B$ is an isomorphism $V \to \mathbb{F}^n$. A vector and its coordinate column are related by a choice of basis; they should not be equated without an appropriate identification.

MOTIVATION AND GUIDING IDEA

Tutorial objectives. Recognize linear maps, compute kernels and images, and use coordinates to pass between polynomials, matrices and vectors. Distinguish a mathematical object from its representation in a chosen basis.

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10.3 Linear maps

DEFINITION 10.4

Let V, W be vector spaces over the same field $\mathbb{F}$.
A map $T : V \to W$ is **linear** if, for all $v_1, v_2 \in V$ and $\lambda \in \mathbb{F}$,

1. $T(v_1 + v_2) = T(v_1) + T(v_2)$.
2. $T(\lambda v_1) = \lambda T(v_1)$.

Interpretation A linear map sends vectors from one vector space to another while preserving their linear structure. It respects vector addition and scalar multiplication.

The first condition says that adding and then applying the map gives the same result as applying the map and then adding. Thus addition commutes with applying the map.

The second condition says that multiplying by a scalar before applying the map gives the same result as multiplying the image by that scalar. In a real geometric setting, scaling an input by λ scales its image by the same factor.

NOTE

Together these properties imply that, for all $\alpha, \beta \in \mathbb{F}$ and $v_1, v_2 \in V$,

$$T(\alpha v_1 + \beta v_2) = \alpha T(v_1) + \beta T(v_2),$$

so a linear map preserves every linear combination.
Geometrically, real linear maps include scalings, rotations and reflections. They send a line to a line or a point and fix the zero vector. In particular, $T(0_V) = 0_W$ always holds.
The main consequences are:

- More generally, $T(\sum_{i=1}^{n} \lambda_i v_i) = \sum_{i=1}^{n} \lambda_i T(v_i)$.
- A necessary condition for $T : V \to W$ to be linear is $T(0_V) = 0_W$. (A map with $T(0_V) \neq 0_W$ is not linear.)

EXERCISE 10.5

$$V = \mathbb{R}_{\leq n}[X], \qquad n \geq 1.$$
$$T_1(P) = P(X) + 2P(-1), \qquad T_2(P) = X^2 P(X),$$
$$T_3(P) = P(X + 2) - P(2).$$

Please decide which formulas define linear maps $V \to V$.

WORKED SOLUTION

T_1 and T_3 do not increase polynomial degree, so they are well-defined and linear. The second formula does not define a self-map of V, since it sends X^n outside V.

$$T_1(\alpha P + \beta Q) = \alpha P + \beta Q + 2\alpha P(-1) + 2\beta Q(-1) = \alpha T_1(P) + \beta T_1(Q).$$

$$T_2(X^n) = X^{n+2} \notin V.$$

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EXERCISE 10.5 CONTINUED

$$T_3(\alpha P + \beta Q) = \alpha P(X+2) + \beta Q(X+2) - \alpha P(2) - \beta Q(2) = \alpha T_3(P) + \beta T_3(Q).$$

MOTIVATION AND GUIDING IDEA

Two guiding questions. The kernel records information lost by a map; the image records attainable outputs. A trivial kernel means that distinct inputs remain distinguishable. An image equal to the codomain means that every output in that codomain is attainable.

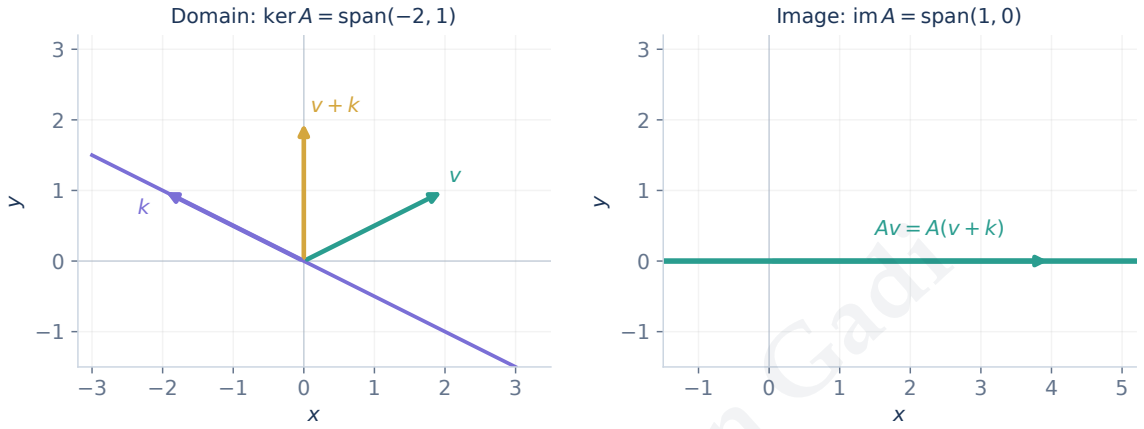


Figure 10.1. The kernel lies in the domain and the image in the codomain. Several inputs can have the same output when the kernel is nontrivial.

10.4 The kernel and image of a linear map

RECALL AND TOOLS

Let $T : V \to W$ be linear.
The **kernel of** T consists of all vectors sent to 0_W:

$$\ker(T) = \{v \in V | T(v) = 0\}$$

The kernel is a subspace of V.
The **image of** T is the set of all outputs attained by the map:

$$\mathrm{Im}(T) = \{T(v) | v \in V\} = \{w \in W | \exists v \in V, T(v) = w\}$$

The image is a subspace of W. It consists of outputs actually attained, whereas the codomain W is specified as part of the definition of the map. These need not be equal.
The sets $\ker T$ and $\mathrm{Im}\, T$ are nonempty, since $T(0_V) = 0_W$.

- Always $\mathrm{Im}\, T \subseteq W$. If $\mathrm{Im}\, T = W$, the map is **surjective**, or **onto**.
- T is **injective**, or **one-to-one**, if, for all $v_1, v_2 \in V$,

  $$T(v_1) = T(v_2) \implies v_1 = v_2$$

  Equivalently, no two distinct inputs have the same image.

  $$v_1 \neq v_2 \implies T(v_1) \neq T(v_2)$$

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- A linear map T is an **isomorphism** if it is injective and surjective.
- Vector spaces V, W are **isomorphic** if there **exists** an isomorphism $T : V \to W$. We then write $V \cong W$.

PROPOSITION 10.2

A linear map $T : V \to W$ is **injective** if and only if $\ker T = \{0_V\}$.
For a homogeneous linear system, this means that the trivial solution is the only solution.

PROPOSITION 10.3

Let $T : V \to W$ be a linear map between finite-dimensional vector spaces. Then

1. If T is injective, $\dim V \leq \dim W$.
2. If T is surjective, $\dim W \leq \dim V$.
3. If $\dim V = \dim W$, then T is injective **if and only if** it is surjective.

EXERCISE 10.6

Let $n \geq 2$ and $V = M_n(\mathbb{F})$. Define $T : V \to \mathbb{F}$ by

$$T(A) = \mathrm{Tr}(A).$$

1. Please prove that T is linear.
2. Please find $\ker T$.
3. Please find $\mathrm{Im}\, T$.
4. Please compute $\dim \ker T$ and $\dim \mathrm{Im}\, T$, and please verify rank-nullity.

WORKED SOLUTION

Write $A = (a_{ij}), B = (b_{ij})$.

1. Linearity:

$$T(A+B) = \mathrm{Tr}(A+B) = \sum_{i=1}^{n}(a_{ii}+b_{ii}) = \sum_{i=1}^{n} a_{ii} + \sum_{i=1}^{n} b_{ii} = T(A) + T(B),$$

For every $\lambda \in \mathbb{F}$,

$$T(\lambda A) = \mathrm{Tr}(\lambda A) = \sum_{i=1}^{n} \lambda a_{ii} = \lambda \sum_{i=1}^{n} a_{ii} = \lambda T(A).$$

Thus T is linear.

2. Kernel:

$$\ker(T) = \{A \in M_n(\mathbb{F}) \ : \ \mathrm{Tr}(A) = 0\}.$$

This is the subspace of matrices with trace zero.

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EXERCISE 10.6 CONTINUED

3. Image: for any $\alpha \in \mathbb{F}$, take

$$A = \operatorname{diag}(\alpha, 0, \ldots, 0) \in M_n(\mathbb{F}),$$

Then $\operatorname{Tr}(A) = \alpha$, so

$$\operatorname{Im}(T) = \mathbb{F}.$$

4. Dimensions: $\dim M_n(\mathbb{F}) = n^2$. Since $\operatorname{Im} T = \mathbb{F}$, its dimension is 1. Rank-nullity therefore gives

$$\dim \ker(T) = n^2 - 1.$$

Alternatively, $n^2 - 1$ entries may be chosen freely, and the last diagonal entry is fixed by requiring trace zero.

EXERCISE 10.7

$$V = C[-1, 2], \qquad S : V \to \mathbb{R}, \qquad S(f) = \int_{-1}^{2} f(x)\, dx.$$

Please prove that S is linear. Please find its kernel and image and please give a nonzero kernel element.

WORKED SOLUTION

Linearity of integration proves linearity. Every real value is obtained from a suitable constant function. The function $x - \frac{1}{2}$ is nonzero and lies in the kernel.

$$S(\alpha f + \beta g) = \int_{-1}^{2} (\alpha f + \beta g) = \alpha \int_{-1}^{2} f + \beta \int_{-1}^{2} g = \alpha S(f) + \beta S(g).$$

$$\ker S = \left\{ f \in C[-1, 2] : \int_{-1}^{2} f(x)\, dx = 0 \right\}, \qquad \operatorname{Im} S = \mathbb{R}.$$

$$S(c/3) = \frac{c}{3}(2 - (-1)) = c, \qquad S(x - \tfrac{1}{2}) = \left[\tfrac{x^2}{2} - \tfrac{x}{2} \right]_{-1}^{2} = 1 - 1 = 0.$$

EXERCISE 10.8

Let V be a vector space over a field F, and let $T : V \to V$ be linear. Please prove:
(a) $\ker T \subseteq \ker T^2$; (b) $\operatorname{Im} T^2 \subseteq \operatorname{Im} T$.

WORKED SOLUTION

(a) We must show that every vector in the set on the left also belongs to the set on the right. Let $v \in \ker T$. Then $T^2(v) = T(T(v)) = T(0) = 0$, so $v \in \ker T^2$.
(b) Let $w \in \operatorname{Im} T^2$. Some $v \in V$ satisfies $w = T^2(v) = T(T(v))$.
Set $u = T(v)$. Then $w = T(u)$, so $w \in \operatorname{Im} T$.

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DEFINITION 10.5

If $T : V \to W$ is an **isomorphism**, each $w \in W$ has a unique $v \in V$ with $T(v) = w$. Thus the **inverse map** $T^{-1} : W \to V$ is defined by $T^{-1}(w) = v$ whenever $T(v) = w$. Applying T to v gives w; applying T^{-1} to w recovers v.

EXERCISE 10.9

$$V = M_2(\mathbb{R}), \qquad \Phi\begin{pmatrix} a & b \\ c & d \end{pmatrix} = (a, b, c, d)^T.$$

$$U = \{x \in \mathbb{R}^4 : x_2 = x_3\}, \qquad W = \{x \in \mathbb{R}^4 : x_1 = x_4 = 0,\ x_2 = -x_3\}.$$

Please prove that Φ is an isomorphism. Please find $\Phi^{-1}(U)$ and $\Phi^{-1}(W)$ and their dimensions.

WORKED SOLUTION

Listing matrix entries preserves addition and scalar multiplication; the inverse formula proves bijectivity. The preimages are the symmetric and skew-symmetric matrices. Their displayed free parameters give the independent spanning matrices and the dimensions.

$$\Phi^{-1}((a, b, c, d)^T) = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

$$\Phi^{-1}(U) = \left\{\begin{pmatrix} a & b \\ b & d \end{pmatrix} : a, b, d \in \mathbb{R}\right\} = \operatorname{span}\left\{\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}\right\}.$$

$$\Phi^{-1}(W) = \left\{\begin{pmatrix} 0 & b \\ -b & 0 \end{pmatrix} : b \in \mathbb{R}\right\} = \operatorname{span}\left\{\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}\right\}.$$

$$\dim U = 3, \qquad \dim W = 1.$$

EXERCISE 10.10 FOUNDATION

Please solve over $\mathbb{R}$ the system

$$\begin{cases} x + y + z = 6, \\ x - y + z = 2. \end{cases}$$

Please write its solution set as a particular solution plus multiples of a fixed vector. Please determine whether it is a subspace and please describe all differences between pairs of solutions.

WORKED SOLUTION

Subtracting the second equation from the first gives $2y = 4$, so $y = 2$ and $x + z = 4$. Put $z = t$ to get

$$S = \{(4 - t, 2, t) : t \in \mathbb{R}\} = (4, 2, 0) + \operatorname{span}\{(-1, 0, 1)\}.$$

It is not a subspace, because zero does not satisfy the equations. The difference of two solutions is

$$(4 - t, 2, t) - (4 - s, 2, s) = (t - s)(-1, 0, 1).$$

Every real multiple occurs by choosing $s = 0$. Thus the differences form exactly

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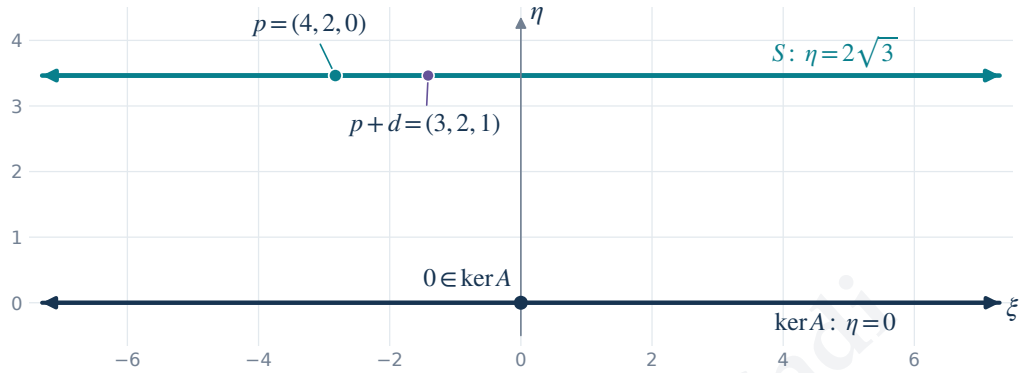
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EXERCISE 10.10 FOUNDATION CONTINUED

$\operatorname{span}\{(-1,0,1)\}$, the solution set of the homogeneous system.

$$S=\{(4-t,2,t):t\in\mathbb{R}\}=p+\ker A$$

$$\Phi(\xi,\eta)=\xi u+\eta v,\qquad u=\frac{(-1,0,1)}{\sqrt{2}},\quad v=\frac{(1,1,1)}{\sqrt{3}}$$



$$\ker A=\operatorname{span}\{d\},\quad d=(-1,0,1),\qquad 0\notin S$$

Figure 10.2. The real system $x+y+z=6$, $x-y+z=2$ has solution set $S=p+\ker A$, where $p=(4,2,0)$ and $\ker A=\operatorname{span}\{(-1,0,1)\}$. The ξ,η axes describe the plane $\operatorname{span}\{u,v\}\subseteq\mathbb{R}^3$ through the map Φ displayed in the figure. The vectors u,v are perpendicular unit vectors, so the chart preserves lengths and angles. The translated line excludes the origin, whereas the kernel contains it.

EXERCISE 10.11 CHALLENGE

Let $a,b\in\mathbb{R}$. Please classify the solutions of

$$\begin{cases}x+y+z=5,\\ x+ay+z=b,\\ x+y+az=b.\end{cases}$$

In each case, please find the coefficient matrix rank, the number of free variables when consistent, and the solution set. Also find the kernel of the coefficient matrix, and please determine whether the solution set is a subspace.

WORKED SOLUTION

Subtract the first equation from both the second and the third. The equivalent system is

$$\begin{cases}x+y+z=5,\\ (a-1)y=b-5,\\ (a-1)z=b-5.\end{cases}$$

1. If $a\neq 1$, division by $a-1$ is permitted. Put $c=(b-5)/(a-1)$. The unique solution is

$$S=\{(5-2c,c,c)\}.$$

There are three pivots, so the rank is 3 and there are no free variables. In the homoge-

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EXERCISE 10.11 CHALLENGE CONTINUED

neous system, $y = z = 0$ and then $x = 0$, so $\ker A = \{0\}$.

2. If $a = 1$ and $b = 5$, the last two equations after subtraction are $0 = 0$. The rank is 1, and there are two free variables. Choosing $y = s, z = t$ gives

$$S = \{(5 - s - t, s, t) : s, t \in \mathbb{R}\} = (5, 0, 0) + \text{span}\{(-1, 1, 0), (-1, 0, 1)\}.$$

The kernel is $\text{span}\{(-1, 1, 0), (-1, 0, 1)\}$.

3. If $a = 1$ and $b \neq 5$, the equation $0 = b - 5$ is a contradiction, so $S = \varnothing$. The coefficient matrix still has rank 1, while the augmented matrix has rank 2. The kernel depends only on A, so it is the same kernel as in the preceding case.

The solution set is never a subspace: in the consistent cases the zero vector fails to satisfy $x + y + z = 5$, and in the inconsistent case the set is empty. A number of degrees of freedom of a solution set cannot be defined through a choice of solution parameters when no solutions exist.

DEFINITION 10.6

Let $\mathbb{F}$ be a field and $A \in \mathbb{F}^{m \times n}$. Write its columns as

$$A = \begin{pmatrix} | & & | \\ a_1 & \dots & a_n \\ | & & | \end{pmatrix}, \qquad a_j \in \mathbb{F}^m.$$

Let $v = \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix} \in \mathbb{F}^n$.

1. **A linear combination of columns.** Matrix-vector multiplication is defined by

$$Av = v_1 a_1 + \cdots + v_n a_n.$$

Thus Av is the linear combination of the columns of A with coefficients $v_1, \dots, v_n$. It follows that

$$\{Av \mid v \in \mathbb{F}^n\} = \text{span}\{a_1, \dots, a_n\} = \text{Col}(A).$$

2. **A linear map.** The function

$$T_A : \mathbb{F}^n \to \mathbb{F}^m, \qquad T_A(v) = Av,$$

is linear: for all $u, v \in \mathbb{F}^n$ and $\lambda \in \mathbb{F}$,

$$T_A(u + v) = A(u + v) = Au + Av,$$

$$T_A(\lambda v) = A(\lambda v) = \lambda(Av).$$

Thus multiplication by A is a linear map, and its columns describe its action on the standard basis of $\mathbb{F}^n$:

$$T_A(e_j) = a_j.$$

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10.5 Determining a map from vector images

PROPOSITION 10.4

Let V, W be vector spaces over the same field $\mathbb{F}$, let $B = (v_1, \dots, v_n)$ be an ordered basis of V, and let $w_1, \dots, w_n \in W$ be arbitrary vectors, not necessarily independent. There is a unique linear map $T : V \to W$ with $T(v_i) = w_i$ for every i. Explicitly, if $v = \sum_{i=1}^{n} a_i v_i$, then

$$T(v) = \sum_{i=1}^{n} a_i w_i.$$

The formula is well defined because basis expansions are unique. It is linear because coordinates of sums and scalar multiples are computed entry by entry. Any linear map with the prescribed images must obey this formula, proving uniqueness. In particular, between any two vector spaces over the same field there is at least one linear map: the zero map.

EXERCISE 10.12

(a) $T(1,0,0)^T = (2,0,1)^T$, $T(1,1,0)^T = (1,1,0)^T$, $T(1,0,1)^T = (0,2,1)^T$;
(b) $T(2,1,4)^T = (1,2,0)^T$, $T(1,0,2)^T = (1,-1,1)^T$, $T(0,1,0)^T = (-1,4,0)^T$.

Please decide in each case whether a linear map $T : \mathbb{R}^3 \to \mathbb{R}^3$ satisfies the requirements. If it exists, please find an explicit formula.

WORKED SOLUTION

In part (a), the three inputs form a basis, so the map exists uniquely. Computing the images of the standard basis gives the formula. In part (b), the proposed outputs do not preserve the displayed input relation, so no such map exists.

(a) $T(e_1) = (2,0,1)^T$, $T(e_2) = (1,1,0)^T - T(e_1) = (-1,1,-1)^T$,

$T(e_3) = (0,2,1)^T - T(e_1) = (-2,2,0)^T$, $T(x,y,z)^T = (2x - y - 2z, y + 2z, x - y)^T$.

(b) $(2,1,4)^T = 2(1,0,2)^T + (0,1,0)^T$,

$2(1,-1,1)^T + (-1,4,0)^T = (1,2,2)^T \neq (1,2,0)^T$.

MOTIVATION AND GUIDING IDEA

Existence criterion for prescribed images. The requirements $T(u_i) = w_i$ can be satisfied by a linear map if and only if, for every choice of coefficients a_i,

$$\sum_i a_i u_i = 0 \quad \Longrightarrow \quad \sum_i a_i w_i = 0.$$

Under this condition, define a map on $U = \operatorname{span}\{u_i\}$ by $T(\sum_i a_i u_i) = \sum_i a_i w_i$. The difference of two representations is a dependence relation, so both yield the same image. Linearity follows from the definition. If the domain is finite-dimensional, extend a basis of U to a basis of V, and freely prescribe images of the extra basis vectors. The map is unique if $U = V$. If $U \neq V$ and $W \neq \{0\}$, two different images may be chosen for an extra basis vector, so the extension is not unique. If the u_i already form a basis, there are no nontrivial dependence

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relations, and every choice of images defines a unique linear map.

EXERCISE 10.13

$$T:\mathbb{R}^2\to\mathbb{R}_3[X],\qquad T((1,0)^T)=1+2X^2,\quad T((1,1)^T)=4+X+2X^2.$$

Please find the unique linear map satisfying these requirements.

WORKED SOLUTION

The two given inputs form a basis. Linearity forces the formula, and substitution reproduces both prescribed outputs.

$$T((0,1)^T)=T((1,1)^T)-T((1,0)^T)=3+X,$$

$$T((x,y)^T)=x(1+2X^2)+y(3+X)=(x+3y)+yX+2xX^2.$$

EXERCISE 10.14

$$P:\mathbb{R}^4\to\mathbb{R}^4,\qquad P(x)=x-\frac{x_1+x_2+x_3+x_4}{4}(1,1,1,1)^T.$$

Please prove that P is linear, please find its kernel and image, and please prove $P^2=P$.

WORKED SOLUTION

The image lies in the zero-sum subspace. Every vector in that subspace is fixed by P, proving equality with the image. The kernel and image dimensions are $1,3$, as rank-nullity requires.

$$\mu(x)=\frac{x_1+x_2+x_3+x_4}{4},$$

$$\begin{aligned}P(\alpha x+\beta y)&=\alpha x+\beta y-(\alpha\mu(x)+\beta\mu(y))(1,1,1,1)^T\\&=\alpha P(x)+\beta P(y).\end{aligned}$$

$$P(x)=0\iff x=\mu(x)(1,1,1,1)^T,\qquad \ker P=\operatorname{span}\{(1,1,1,1)^T\}.$$

$$\sum_{j=1}^{4}(Px)_j=\sum_{j=1}^{4}x_j-4\mu(x)=0.$$

$$\operatorname{Im}P=\{y\in\mathbb{R}^4:y_1+y_2+y_3+y_4=0\}=\operatorname{span}\{e_1-e_4,e_2-e_4,e_3-e_4\}.$$

$$\mu(Px)=0\quad\Longrightarrow\quad P(Px)=Px.$$

Additional practice

EXERCISE 10.15 CHALLENGE

$$P:V\to V,\quad P^2=P.$$

$$V=\ker P\oplus\operatorname{Im}P,\quad \ker(I-P)=\operatorname{Im}P,\quad \operatorname{Im}(I-P)=\ker P.$$

$$P_0(x,y,z)^T=(x+2z,y-z,0)^T\quad\text{on }\mathbb{R}^3.$$

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EXERCISE 10.15 CHALLENGE CONTINUED

Please prove the three identities and the existence and uniqueness of the projection with kernel U and image W when $V = U \oplus W$. Please compute the explicit decomposition for P_0.

WORKED SOLUTION

The first decomposition splits a vector into a kernel part and an image part. The map acts as the identity on its image, so the intersection is zero. The displayed calculations prove all three identities. For a given $U \oplus W$, uniqueness of decomposition makes $P(u+w) = w$ well-defined and linear. It is the only possibility because it must vanish on the kernel and fix the image. In the example, the two parts lie in the stated subspaces.

$$v = (v - Pv) + Pv, \quad P(v - Pv) = 0, \quad P(Pv) = Pv.$$

$$w \in \ker P \cap \operatorname{Im} P \implies Pw = 0 = w \implies w = 0.$$

$$(I - P)v = 0 \iff Pv = v \iff v \in \operatorname{Im} P, \quad P(I - P) = 0, \quad u \in \ker P \implies (I - P)u = u.$$

$$P(u + w) = w \quad (u \in U, w \in W),$$

$$\ker P_0 = \operatorname{span}\{(-2, 1, 1)^T\}, \quad \operatorname{Im} P_0 = \{(a, b, 0)^T : a, b \in \mathbb{R}\},$$

$$(x, y, z)^T = z(-2, 1, 1)^T + (x + 2z, y - z, 0)^T.$$

10.6 From a Spanning Description to Linear Equations

RECALL AND TOOLS

Let $U \leq \mathbb{F}^n$. If the columns of C form a basis of U, then

$$x \in U \iff Ct = x \text{ is consistent.}$$

Apply row operations to $[C \mid x]$, leaving the entries of x as variables. If $\dim U = d$, the reduced form of C is $\begin{pmatrix} I_d \\ 0 \end{pmatrix}$. The bottom rows of the augmented matrix give $n - d$ homogeneous equations in the entries of x. Place their coefficients in a matrix A to obtain $U = \ker A$.
Conversely, row reduction of A expresses all solutions of $Ax = 0$ in terms of the free variables. Setting one free variable to 1 and the others to 0 gives a basis of the kernel. Here $\ker A = \{x : Ax = 0\}$.
If $d = n$, there are no equations; the $1 \times n$ zero matrix is a possible choice. If $d = 0$, then $U = \{0\} = \ker I_n$.

EXAMPLE 10.1

Over a field $\mathbb{F}$, let

$$U = \operatorname{span}\left\{ \begin{pmatrix} 1 \\ 0 \\ 2 \\ -1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \\ 3 \end{pmatrix} \right\}.$$

The two columns are independent, as their first two entries show. Reduce the augmented

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matrix:

$$\left(\begin{array}{cc|c} 1 & 0 & x_1 \\ 0 & 1 & x_2 \\ 2 & 1 & x_3 \\ -1 & 3 & x_4 \end{array}\right) \longrightarrow \left(\begin{array}{cc|c} 1 & 0 & x_1 \\ 0 & 1 & x_2 \\ 0 & 0 & x_3 - 2x_1 - x_2 \\ 0 & 0 & x_4 + x_1 - 3x_2 \end{array}\right).$$

The operations are $R_3 \leftarrow R_3 - 2R_1 - R_2$ and $R_4 \leftarrow R_4 + R_1 - 3R_2$. Hence

$$U = \left\{ x \in \mathbb{F}^4 : \begin{array}{rcl} -2x_1 - x_2 + x_3 & = & 0, \\ x_1 - 3x_2 + x_4 & = & 0 \end{array} \right\} = \ker \begin{pmatrix} -2 & -1 & 1 & 0 \\ 1 & -3 & 0 & 1 \end{pmatrix}.$$

To recover the spanning description, set $x_1 = s$ and $x_2 = t$. Then $x_3 = 2s+t$ and $x_4 = -s+3t$, so

$$x = s \begin{pmatrix} 1 \\ 0 \\ 2 \\ -1 \end{pmatrix} + t \begin{pmatrix} 0 \\ 1 \\ 1 \\ 3 \end{pmatrix}.$$

The same calculation works over finite fields, interpreting the numbers and operations in the field.

EXERCISE 10.16

Let $A \in M_{r\times n}(\mathbb{F})$ have rank $r \geq 1$, and let $B \in M_{k\times n}(\mathbb{F})$. Please prove that

$$\ker A \subseteq \ker B \quad \Longleftrightarrow \quad \text{there is a unique } C \in M_{k\times r}(\mathbb{F}) \text{ with } B = CA.$$

WORKED SOLUTION

If $B = CA$ and $Ax = 0$, then $Bx = CAx = C0 = 0$. Thus $\ker A \subseteq \ker B$.
Now assume the inclusion. Since $\operatorname{rank} A = r$, each standard column $e_j \in \mathbb{F}^r$ has a preimage $d_j \in \mathbb{F}^n$ satisfying $Ad_j = e_j$. Place these solutions in the columns of

$$D = (d_1 \ \cdots \ d_r) \in M_{n\times r}(\mathbb{F}), \qquad AD = I_r.$$

For every $x \in \mathbb{F}^n$,

$$A(x - DAx) = Ax - (AD)Ax = Ax - Ax = 0.$$

The inclusion gives $B(x - DAx) = 0$, hence $Bx = BDAx$. This identity holds for every x, so

$$B = BDA.$$

Therefore $C = BD$ works. If also $B = C'A$, then

$$BD = C'AD = C'I_r = C'.$$

Every admissible matrix must therefore equal BD, proving uniqueness.

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EXERCISE 10.17

Let $U \leq \mathbb{F}^n$ have dimension $1 \leq d \leq n$. Please prove that d standard coordinates can be selected so that their values determine each vector in U uniquely, and every choice of d values occurs. Please prove that fewer than d coordinates cannot suffice.

WORKED SOLUTION

Choose a basis $u_1, \ldots, u_d$ of U and place it in the columns of

$$C = (u_1 \ \cdots \ u_d) \in M_{n\times d}(\mathbb{F}).$$

The rank of C is d. Thus d independent rows can be selected. Write their indices as $j_1 < \cdots < j_d$, and let R be the square matrix consisting of these rows. It has rank d and is invertible. Every $x \in U$ has a unique expression $x = Ct$. The selected coordinates satisfy

$$\begin{pmatrix} x_{j_1} \\ \vdots \\ x_{j_d} \end{pmatrix} = Rt.$$

If their prescribed values form a column $y \in \mathbb{F}^d$, then

$$t = R^{-1}y, \qquad x = CR^{-1}y.$$

This is a unique vector of U, and every y is possible.
If only $k < d$ coordinates are selected, the corresponding rows of C form a $k \times d$ matrix. Its homogeneous system has a solution $t \neq 0$. The columns of C are independent, so $Ct \neq 0$, but all k selected coordinates of Ct vanish. Those coordinates do not distinguish Ct from the zero vector.

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EXERCISE 10.18 CHALLENGE

Let V have finite dimension $n \geq 2$ over a field F. Let $T : V \to V$ be linear and satisfy $T(v) \in \text{span}\{v\}$ for every $v \in V$. Please prove that one scalar λ satisfies $T(v) = \lambda v$ for all v. Please check whether imposing the condition only on a fixed basis is sufficient.

WORKED SOLUTION

Choose a basis $e_1, \ldots, e_n$ and write $T(e_j) = \lambda_j e_j$. For $i \neq j$, the hypothesis for $e_i + e_j$ gives

$$\lambda_i e_i + \lambda_j e_j = T(e_i + e_j) = \mu(e_i + e_j)$$

for some μ. Independence implies $\lambda_i = \lambda_j = \mu$. All basis scalars therefore equal one λ, and linearity gives

$$T\left(\sum_j a_j e_j\right) = \sum_j a_j \lambda e_j = \lambda \sum_j a_j e_j.$$

A basis condition alone is insufficient. On $\mathbb{R}^2$, $T(x, y) = (x, 2y)$ preserves the lines of the two standard basis vectors, but $T(1, 1) = (1, 2)$ is not a multiple of $(1, 1)$.

EXERCISE 10.19 EXAM QUESTION

1. Let $n \geq 2$ and $A, B \in M_n(\mathbb{R})$ satisfy $B \neq 0$ and $\text{rank}(A) = n - 1$. Please prove that $AB = 0$ implies $\text{rank}(B) = 1$.

2. Let $A, B \in M_{2031}(\mathbb{R})$ satisfy $AB = 0$. Please prove that

$$\text{rank}(A) + \text{rank}(B) \leq 2031.$$

WORKED SOLUTION

The key implication in both parts is

$$AB = 0 \quad \Longrightarrow \quad \text{Im}(B) \subseteq \ker A.$$

Indeed, every $y \in \text{Im}\, B$ has the form $y = Bx$. Thus $Ay = A(Bx) = (AB)x = 0$, so $y \in \ker A$.

1. The inclusion and rank-nullity give

$$\begin{aligned} \text{rank}(B) &= \dim \text{Im}(B) \\ &\leq \dim \ker A \\ &= n - \text{rank}(A) \\ &= n - (n - 1) = 1. \end{aligned}$$

   Since $B \neq 0$, at least one column of B is nonzero, so $\text{rank}(B) \geq 1$. The two bounds yield $\text{rank}(B) = 1$.

2. Apply the same inclusion in $\mathbb{R}^{2031}$:

$$\begin{aligned} \text{rank}(B) &= \dim \text{Im}(B) \\ &\leq \dim \ker A \\ &= 2031 - \text{rank}(A). \end{aligned}$$

   Rearranging gives

$$\text{rank}(A) + \text{rank}(B) \leq 2031.$$

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EXERCISE 10.19 EXAM QUESTION CONTINUED

If every output of B is annihilated by A, the entire image of B must lie in the kernel of A. The dimension of that kernel therefore bounds the rank of B.

EXERCISE 10.20

$$A = \begin{pmatrix} 1 & 0 & -3 \\ 0 & 1 & 2 \end{pmatrix}, \qquad z = ax_1 + bx_2 + cx_3.$$

Please find a necessary and sufficient condition for uniquely recovering $x \in \mathbb{R}^3$ from Ax and z. Please give one useful extra reading and one reading determined by the original two.

WORKED SOLUTION

Unique recovery means that the extra reading does not vanish on the kernel direction. Equivalently, the augmented square matrix is invertible. The first example satisfies the condition; the second is the sum of the two original readings.

$$\ker A = \operatorname{span}\{(3,-2,1)^T\}, \qquad (a,b,c)(3,-2,1)^T = 3a - 2b + c.$$

$$\det \begin{pmatrix} 1 & 0 & -3 \\ 0 & 1 & 2 \\ a & b & c \end{pmatrix} = 3a - 2b + c.$$

$$3a - 2b + c \neq 0.$$

$$z = x_3 : \quad (a,b,c) = (0,0,1), \qquad z = x_1 + x_2 - x_3 : \quad (a,b,c) = (1,1,-1).$$

EXERCISE 10.21 FOUNDATION

For $t \in \mathbb{R}$, consider

$$A_t = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 4 & 1 \\ 3 & 6 & t \end{pmatrix}.$$

1. Please compute $\det A_t$ using one row operation, and please determine when A_t is invertible.
2. At $t = 0$, please find a basis of $\ker A_0$ and please find $\operatorname{rank} A_0$.
3. Please construct B_t from A_t by, in this order, interchanging the first two rows, multiplying the third row by 3, and adding 5 times the third row to the first row. Please compute $\det B_t$ without writing B_t explicitly.

WORKED SOLUTION

1. The operation $R_3 \leftarrow R_3 - 3R_1$ preserves the determinant and gives

$$\begin{pmatrix} 1 & 2 & 0 \\ 0 & 4 & 1 \\ 0 & 0 & t \end{pmatrix}.$$

Hence $\det A_t = 1 \cdot 4 \cdot t = 4t$, and A_t is invertible exactly when $t \neq 0$.

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EXERCISE 10.21 FOUNDATION CONTINUED

2. At $t = 0$, the kernel equations are $x_1 + 2x_2 = 0$ and $4x_2 + x_3 = 0$. Setting $x_2 = s$ gives

$$\ker A_0 = \{s(-2, 1, -4)^T : s \in \mathbb{R}\} = \text{span}\{(-2, 1, -4)^T\}.$$

The kernel has dimension 1, so rank-nullity gives $\text{rank } A_0 = 3 - 1 = 2$.

3. The interchange multiplies the determinant by -1, scaling the row multiplies it by 3, and the final operation preserves it. Thus

$$\det B_t = (-1) \cdot 3 \cdot \det A_t = -12t.$$

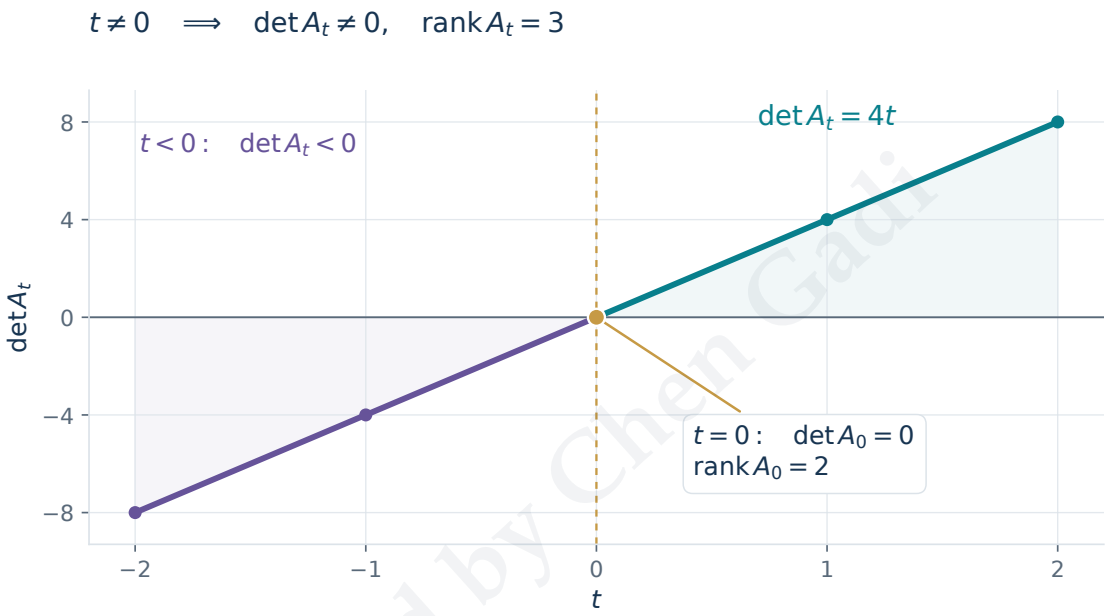


Figure 10.3. For the matrix A_t in the exercise, the graph $\det A_t = 4t$ crosses zero only at $t = 0$. For every $t \neq 0$, the matrix is invertible and has rank 3; at the gold point, $\text{rank } A_0 = 2$, with kernel $\text{span}\{(-2, 1, -4)^T\}$.

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EXERCISE 10.22 CHALLENGE

Let $A \in M_n(\mathbb{F})$ be invertible, $U \in \mathbb{F}^{n \times k}$, and $V \in \mathbb{F}^{k \times n}$, with $n, k \geq 1$.

1. Please use determinants of block matrices to prove

$$\det(A + UV) = \det A \, \det(I_k + VA^{-1}U).$$

Do not assume that U or V is square or invertible.

2. For $s \in \mathbb{R}$, please use the identity to compute the determinant of

$$C_s = \begin{pmatrix} s+2 & 0 & 1 \\ 0 & s+2 & -1 \\ s+1 & s+1 & 1 \end{pmatrix}$$

and please determine when it is invertible. At the singular parameter value, please find its kernel and rank.

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EXERCISE 10.22 CHALLENGE CONTINUED

WORKED SOLUTION

1. First prove $\det(I_n + XY) = \det(I_k + YX)$ for $X \in \mathbb{F}^{n\times k}$ and $Y \in \mathbb{F}^{k\times n}$. Set

$$K = \begin{pmatrix} I_n & X \\ -Y & I_k \end{pmatrix}, \quad L = \begin{pmatrix} I_n & 0 \\ Y & I_k \end{pmatrix}, \quad H = \begin{pmatrix} I_n & -X \\ 0 & I_k \end{pmatrix}.$$

Both L and H are block triangular with determinant 1. We have

$$LK = \begin{pmatrix} I_n & X \\ 0 & I_k + YX \end{pmatrix}, \qquad HK = \begin{pmatrix} I_n + XY & 0 \\ -Y & I_k \end{pmatrix}.$$

Multiplicativity and the determinant formula for block triangular matrices give

$$\det K = \det(I_k + YX) = \det(I_n + XY).$$

Now $A + UV = A(I_n + A^{-1}UV)$. Substitute $X = A^{-1}U$ and $Y = V$ to obtain the required identity.

2. Take $A = I_3$ and write $C_s = I_3 + UV$, where

$$U = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix}, \qquad V = \begin{pmatrix} s+1 & 0 & 1 \\ 0 & s+1 & -1 \end{pmatrix}.$$

Direct multiplication verifies this factorization and gives

$$I_2 + VU = \begin{pmatrix} s+3 & 1 \\ -1 & s+1 \end{pmatrix}.$$

Therefore

$$\det C_s = \det(I_2 + VU) = (s+1)(s+3) + 1 = (s+2)^2.$$

Thus C_s is invertible exactly when $s \neq -2$. At $s = -2$, its kernel equations are $x_3 = 0$ and $-x_1 - x_2 + x_3 = 0$, so

$$\ker C_{-2} = \operatorname{span}\{(1, -1, 0)^T\}, \qquad \operatorname{rank} C_{-2} = 3 - 1 = 2.$$

The multiplicity of a root of the determinant as a polynomial in s is not the kernel dimension: this root is double, while the kernel has dimension 1.

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EXERCISE 10.23

Let $A \in M_n(\mathbb{F})$ with $n \geq 2$.

1. Please prove that $\operatorname{adj}(A)$ has rank n, 1, or 0 according as rank $A = n$, rank $A = n-1$, or rank $A \leq n-2$.
2. Please deduce that $\det(\operatorname{adj}(A)) = (\det A)^{n-1}$.

WORKED SOLUTION

1. We use the rank characterization by minors. If a matrix has rank r, select r independent columns. The resulting matrix has row rank r, so selecting r independent rows gives an invertible $r \times r$ minor. Every larger minor has dependent columns and determinant zero. If A is invertible, then $\operatorname{adj}(A) = (\det A)A^{-1}$ is invertible and has rank n. If rank $A = n-1$, some minor of order $n-1$ is nonzero. At least one cofactor is therefore nonzero, so $\operatorname{adj}(A) \neq 0$. On the other hand,

$$A \operatorname{adj}(A) = 0,$$

so every column of the adjugate belongs to $\ker A$. Rank-nullity gives

$$\dim \ker A = n - (n-1) = 1.$$

Hence $1 \leq \operatorname{rank}(\operatorname{adj}(A)) \leq 1$, and the rank is 1. If rank $A \leq n-2$, every minor of order $n-1$ vanishes. Every cofactor is zero, so $\operatorname{adj}(A) = 0$.

2. For invertible A, take determinants in $A \operatorname{adj}(A) = (\det A)I_n$:

$$\det A \cdot \det(\operatorname{adj}(A)) = (\det A)^n.$$

Division by $\det A \neq 0$ gives the result. If A is singular, part 1 gives $\operatorname{rank}(\operatorname{adj}(A)) \leq 1 < n$, so both sides of the required identity are zero.

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